

Non-Repudiation -- Guided Notes

Unit tp-6.8 | CompTIA Network+ N10-009 Obj. FC0-U71 6.1 | N10-008 FC0-U71 6.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Non-repudiation is the security property that ensures a party _____ deny having sent a message or performed an action after the fact. It provides _____ of origin (sender cannot deny sending) and _____ of receipt (receiver cannot deny receiving). It is achieved through a combination of _____ and _____.

Question 2

Digital signatures provide non-repudiation for electronic documents. The sender uses their _____ key to sign the hash of the document. The receiver uses the sender's _____ key to verify the signature. Only the sender's private key could have produced that signature, so the sender _____ deny signing the document.

Question 3

A Certificate Authority (CA) is a trusted third party that issues _____ certificates that bind a public key to an identity. This is part of the PKI (Public Key Infrastructure). When a CA signs a certificate, it provides non-repudiation for _____ transactions by confirming that the public key genuinely belongs to the stated _____.

Question 4

A digital signature is _____ from a written signature because it is cryptographically bound to the _____ of the document. If the document changes after signing, the signature becomes _____. A written signature _____ detect document tampering after the fact.

Question 5

In legal and financial contexts, non-repudiation ensures that electronic contracts and transactions are _____. The US E-SIGN Act and UETA give electronic signatures the same _____ weight as handwritten signatures. eSignature platforms like DocuSign use PKI-based digital signatures to provide court-admissible _____.

Question 6

Timestamping adds non-repudiation for _____ -- proving that a signature or document existed at a specific point in time. A trusted timestamping authority (TSA) signs a hash of the document along with the current _____, creating a verifiable record that the document existed in its current form at that _____.

Question 7

HMAC (Hash-based Message Authentication Code) provides _____ and data _____, but not non-repudiation. This is because HMAC uses a _____ key -- both sender and receiver have the same key, so either party could have created the MAC. Non-repudiation requires _____ keys (only the sender's private key can sign).

Question 8 -- Real World Application

REAL WORLD: A contractor submits an invoice via email claiming they delivered 200 hours of work. The company disputes receiving the deliverables and refuses to pay. The contractor says they sent the files; the company says they never arrived. There is no digital signature on the email. Explain what non-repudiation mechanism would have resolved this dispute and describe how it works.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-6-8-non-repudiation/tp-6-8-non-repudiation-answer-key.pdf
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